

On the Construction of Configurations n_3
Continuation: Linear Constructions and Conclusion
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Linear Constructions

§8. General problem

We have seen that the construction of any Cf. n_3 can be carried out by a number of linear operations and a single quadratic one. In special cases, however, the task of finding the double elements of two incident point rows, or of two pencils of rays, reduces to linear constructions; in that case we are able to construct the Cf. n_3 in the most general way by straightedge alone. Even when the procedure given in §§6–7 makes a quadratic construction unavoidable, it is not excluded that another procedure may require only linear operations.

If a general linear construction of the Cf. is impossible, it is nevertheless possible in most cases to obtain certain forms of the required Cf. by linear construction. Schroeter first gave such constructions for those Cff. which can be regarded as polygons in which the first, second, and fourth vertices always lie on one line. Schroeter's procedure can also be used for many other Cff.; he himself used it to obtain those Cff. 10_3 for which a general linear construction did not succeed. In what follows, without imposing any special hypothesis on the Cff. under consideration, a procedure will be given by which one arrives at their linear construction. For each Cf. to which this procedure is to be applied, a separate examination is nevertheless needed, since in certain cases the procedure becomes illusory.

§9. The two remaining free elements

In order to arrive at a general construction of the Cff. n_3 , we previously looked in the sequence of its elements

$$p_1, r_1, p_2, r_2, \dots, p_n, r_n$$

for an element that was not yet completely determined by the preceding elements, but that, together with them, determined all the following elements. The element sought is either a point p_i or a line r_i ; for the moment we shall not distinguish the two cases, and shall denote the element by e_i . The case previously avoided, in which e_i remains completely undetermined after the construction of the preceding elements, is favorable for the present purpose; but since it is easily disposed of after the other cases are considered, we exclude it for the moment.

We now look, in the sequence of elements preceding e_i , for an element which is not yet completely determined by the preceding elements, but after whose construction the following elements up to e_i , excluding e_i itself, are uniquely determined. It is clear that there must always be one and only one such element. We call it e_h , and express by the number h that it is identical with either p_h or r_h . After the preceding elements have been constructed, e_h may still be completely undetermined. We again exclude this case temporarily, since it too will be easy to settle later.

The element e_i always belongs to the last polygon, while e_h belongs either to the last or to the next-to-last polygon. If it belongs to the next-to-last, then e_i must be the first point of the last polygon. This property remains valid for the Cf. if we cyclically permute the lines of the last complex. We again make such a cyclic permutation so that p_m , the first point of the last polygon, becomes identical with t_{m-1} .

We now construct the elements preceding e_h by the method of §6, and ask whether and how it is then still possible to complete the Cf. using only the straightedge. We denote by A the totality of elements preceding e_h ,

$$p_1, r_1, p_2, r_2, \dots, p_{h-1}, r_{h-1} \quad (\text{or } p_h),$$

and by B the remaining elements

$$(p_h), r_h, p_{h+1}, r_{h+1}, \dots, p_n, r_n.$$

By our assumptions, e_h is incident with exactly one element of A ; the other two elements incident with e_h therefore belong to B . Of these two, one is always the element immediately following e_h ; the other will be denoted by η^h . Of the elements incident with e_i , exactly one must precede e_i ; of the other two, one immediately follows e_i , and the other will be denoted by η^i .

The elements e_h, e_i, η^h, η^i occur in the sequence

$$p_1, r_1, p_2, r_2, \dots, p_n, r_n$$

either in the order

$$e_h, e_i, \eta^h, \eta^i \quad (\text{Case I}),$$

or in the order

$$e_h, e_i, \eta^i, \eta^h \quad (\text{Case II}),$$

or in the order

$$e_h, \eta^h, e_i, \eta^i \quad (\text{Case III}).$$

It will be shown that in Cases I and II the Cf. can be completed linearly in arbitrarily many different ways, whereas in Case III no linear solution of the problem is possible in general.

The last line r_n contains the three points t_n, p_n, p_m . It may happen that r_n is identical with one of e_h, e_i , while p_m is identical with the other; then $r_n = \eta^h = \eta^i$. Although the construction of the Cf. in this case presents no difficulty, we shall always regard it as avoided. It must be shown that it can always be avoided.

More generally, whenever the present decomposition into complexes contains more than one complex, a mere change in the arrangement of the last complex can ensure that r_n is distinct from e_h and from e_i ; hence, since the point p_m identical with t_n cannot be completely determined after the preceding elements are constructed, it belongs to A . If the number k of complexes is greater than 1, at least one element of R_k , call it $v = p$, must be a vertex of one of the preceding polygons; otherwise C_k alone would form a Cf.. If e_h belongs to the last polygon, then $p = r$ belongs to A , and a cyclic permutation of the columns of C_k can make $p = r$ the first point of the last line. Thus that line belongs to A .

If e_h belongs to the next-to-last polygon, this cyclic permutation is not admissible, because the columns of the last complex were arranged so that $e_i = p_m$ should be identical with t_{m-1} . In this case r_n cannot be identical with e_i , since r_n contains $t_n, p_n, p_m = e_i$; it remains only to treat the case $r_n = e_h$. If this occurs, we interchange the second and third horizontal rows of C_k and reverse the order of the columns of this complex. We then obtain as P_k the same polygon as before, but in the reverse order. After a cyclic permutation of the columns of the last complex so that $e_i = p_m$ is identical with t_{m-1} , the element now denoted by e_i is the same as before, while the point now denoted by t_n is different from the previous one. Hence r_n and e_h are distinct, and t_n belongs to A .

§10. General auxiliary facts

For all cases the following observations apply.

1. The elements e_h, e_i, η^h, η^i are always distinct. Of e_h and η^h , one is a point and the other a line; the same holds for e_i and η^i .

2. Every element of B distinct from e_h, e_i, η^h, η^i is incident with the immediately preceding element, with the immediately following element unless it is r_n , and with an element of A .

Indeed, if e is such an element, it is incident with the immediately preceding element, since every element of the Cf. which is not the first point of a polygon has this property. Among the elements of B , only e_h and e_i can fail to have it. Since e is completely determined by preceding elements, it must be incident with two of them. Let e' be the first element, in the normal order, which is incident with e . Between e' and e there is another element incident with e , namely the element immediately preceding e . The element e' cannot be identical with e_h or e_i , and it cannot be identical with η^h or η^i either. If e' belonged to B , then e' would be incident with its immediate predecessor and, since it is not r_n , also with its immediate successor; together with the incidence with e , all three elements incident with e' would belong to B . Hence, if among the elements of B different from e_h, e_i, η^h, η^i there were an element not incident with any element of A , it would be preceded by another element with the same property. This is impossible, since B contains only finitely many elements. Thus every such e is incident with an element e' of A .

3. The latest among the four elements e_h, e_i, η^h, η^i in the sequence may be r_n . Then one of the points t_n, p_m is identical with one of e_h, e_i , and only one of these points can be so. If $p_m = e_h$ or e_i , then r_n belongs to A ; if $t_n = e_h$ or e_i , then e_h and e_i lie in the last polygon and are distinct from p_m , so that p_m belongs to A .

If η^h and η^i are distinct from r_n , say one of them is p_s or r_s , then the conditions of the Cf., including the last condition expressing the incidence of t_n with r_n , prescribe the construction of the elements following η :

$$(r_s), p_{s+1}, r_{s+1}, \dots, p_n, r_n.$$

The elements $(\eta), p_{s+1}, r_{s+1}, \dots, p_n, r_n$ and also p_m must all belong to A . Hence the successive intersections and joining lines are obtained as before:

$$p_m r_n = r_n, \quad (r_n, p_n) = p_n, \quad p_n r_{n-1} = r_{n-1}, \quad \dots, \quad (r_{s+1}, p_{s+1}) = p_{s+1}.$$

Consequently, after all elements belonging to A have been constructed, all elements following η are determined if the last condition of the Cf. is also taken into account. We construct these elements, denote all elements constructed so far by A' , and the remaining ones by B' . In choosing e_h and e_i we must therefore ensure that the element η determined by them is incident with the immediately following element of A' ; if η^h or η^i is r_n , this following element is replaced by whichever of t_n, p_m belongs to A .

§11. Reduction of the cases

Assume first that Case I or II occurs. Once all elements belonging to A' have been constructed, the remaining elements can, after a completely arbitrary choice of e_i , still be determined so that only one of all the conditions of the Cf. remains unfulfilled. In these two cases none of the elements η^h, η^i lies between e_h and e_i . Every element between e_h and e_i is therefore incident, by §10,2, with its immediate predecessor, its immediate successor, and an element of A . Thus, once e_i is chosen, the elements between e_h and e_i , including e_h itself, are obtained in reverse order, each as an intersection or joining line of the following, already constructed element with an element of A .

By the definition of e_i , all elements following e_i are now determined and can be obtained by the procedure of §6. Those following η have already been constructed; if the remaining ones up to η are constructed, then $3n - 1$ of the $3n$ conditions of the Cf. are satisfied. The remaining condition is that, for an arbitrary choice of e_i , the element η will not in general be incident with the following element. In order for this condition also to hold, e_i must be a point or a tangent of a certain curve. The task is therefore to determine, by linear constructions, arbitrarily many points or tangents of this curve.

Case III is essentially different, because there η^h lies between e_h and e_i and η^h is incident with no element of A' .

In addition to the order in which e_h, e_i, η^h, η^i occur, we must distinguish their nature. Either

- a) $e_h = p_h, \quad e_i = p_i, \quad \eta^h$ and η^i are lines,
- b) $e_h = r_h, \quad e_i = r_i, \quad \eta^h$ and η^i are points,
- c) $e_h = p_h, \quad e_i = r_i, \quad \eta^h$ is a line, η^i a point,
- d) $e_h = r_h, \quad e_i = p_i, \quad \eta^h$ is a point, η^i a line.

Thus there are twelve cases, denoted

Ia	Ib	Ic	Id
IIa	IIb	IIc	IId
IIIa	IIIb	IIIc	IIId.

§12. Case I

In Cases I and II, e_h cannot be the first point of the next-to-last polygon; otherwise $\eta^h = r_{m-1}$, $\eta^i = r_m$, and $e_i = p_m$, so that η^h would occur between e_h and e_i , which is Case III. In Case I, e_h cannot be the first point of the last polygon either, for then $\eta^h = r_n = \eta^i$, giving Case II.

We first treat Case Ia. Then $e_h = p_h, e_i = p_i$, with the corresponding incident elements denoted as above. Suppose the relevant indices satisfy $h < i < j < l$. The elements belonging to A' are already constructed. As shown in §11, after an arbitrary choice of p_i the remaining elements of B' may be constructed so that only the condition requiring the incidence of $p_l = r_1$ and p_{l+1} is not satisfied. One may instead impose that condition and leave another one out.

Following the prescriptions of the Cf., one obtains chains of intersections and joining lines from the chosen point p_i ; for arbitrary p_i the only condition left open is that the points

$$p_h = t_j, \quad p_j, \quad p_{j+1}$$

should lie on one line, namely the Cf.-line r_j .

If the point p_i moves in the plane, then the other elements of B' also move. The lines p_i, r_{i-1}, r_i rotate about fixed points, while the points $p_h = t_j, p_j, p_{j+1}$ run through fixed lines. Each of the corresponding pencils is projectively related to the corresponding point row. Thus the construction is reduced to the following problem.

Given are three points, the centers of three pencils of rays, and three lines, the carriers of three point rows. To each pencil a point row is projectively assigned. Through any point p of the plane there passes one ray of each pencil; to these three rays correspond three points in the assigned point rows. One must find those points p for which the three points thereby obtained lie on one line.

Every solution of this problem also solves the dual problem: find lines in the plane such that the three rays corresponding to their intersections with three given lines meet in a point. This dual problem gives Case Ib. Cases Ic and Id are obtained by the same reasoning. In Case Ic, the moving element is a line r_i rather than a point p_i ; the relevant points move on projectively

related rows, and one is led to this form: given three lines carrying point rows and three further lines carrying point rows projectively assigned to the first three, find lines which cut the first three in points whose assigned points again lie on one line. Case Id gives the dual problem.

§13. Case II

We next treat Case IIa. Here $e_h = p_h$, $e_i = p_i$, while the two elements η^h, η^i occur in the reverse order. With the notation chosen in the text, all relevant elements preceding and following the two free choices belong to A' except the elements to be constructed from p_i . If p_i is chosen arbitrarily in the plane, then it determines all the missing elements by the successive intersection and joining operations prescribed by the Cf.. The only condition neglected is that the point $p_i = r_1$ should lie on the line r_1 .

As p_i varies, two of the determined lines describe pencils and two of the determined points describe point rows projectively related to those pencils. We arrive at the following problem. Given are two points, centers of two pencils, and two lines, carriers of two point rows; each row is projectively related to one of the pencils. These relations assign to each point of the plane a line, and conversely to each such line the same point. Through the point pass two rays, and to those rays correspond two points; their joining line is the line assigned to the given point. One must find those points which lie on their assigned lines. Equivalently, one may ask for those lines which pass through their assigned points. This gives the dual Case IIb.

Case IId differs only superficially from IIa and leads to the same problem; Case IIc is dual to IId.

§14. Case III

In Case IIIa the order is e_h, η^h, e_i, η^i . The construction can be split into three independent projective motions: one point runs through a point row, another point runs through a second point row, and a line rotates in a pencil. Taking into account the two still unfulfilled Cf.-conditions, one obtains the following problem.

Given are two projective point rows, one point row projective to a pencil, and two projective pencils. It is required to determine corresponding pairs of elements such that three points lie on one line and three lines pass through one point. Case IIIb leads to the same problem.

Similarly Case IIIc leads to this problem. Given are three pairs of lines $a, a_1; b, b_1; c, c_1$. Points of a are projectively assigned to points of a_1 , points of b to points of b_1 , and points of c to points of c_1 . One has to find corresponding points $a, a_1; b, b_1; c, c_1$ such that a, a_1, b are collinear and b_1, c, c_1 are collinear. Case IIId gives the dual problem.

§15. The excluded cases

It remains to discuss the two cases excluded in §9. If e_i is not determined at all by the preceding elements, then necessarily $e_i = p_m$. The point t_{m-1} must be a vertex of the last polygon. By cyclically permuting the lines of the last complex so that this vertex becomes the first vertex of the last polygon, and by changing the notation of the vertices and sides of that polygon accordingly, the previous point p_m obtains a later name $p_s = r_t$ with $s, t > m$. One sees that the situation is reduced to one of Cases IIa or IIc.

If e_h is not determined at all by the preceding elements, then it is the first point of the next-to-last or last polygon. If it is the first point of the next-to-last polygon, then a reordering of the complexes placing that complex last reduces the case to the previous one. If the line

involved belongs to the last complex, then after one arbitrary joining line and a chain of the standard intersections and joins, the only remaining condition is again that three corresponding points should be collinear; this is the same problem as in Case Ia.

Finally suppose e_h is the first point of the last polygon and remains completely undetermined. We call this Case IV. Every element of the last polygon is incident with its two neighbors. In the present case e_h is also incident with a non-neighbor, denoted η_h , and e_i with a non-neighbor η_i . Among all vertices and sides of the last polygon, only e_h, e_i, η_h, η_i have the property of also being incident with a non-neighbor. This distinguished set is unchanged by cyclic permutation of the lines of the last complex.

Among these four elements are two points and two lines. Denote the points by e_1, e_2 , and the non-neighboring lines through them by η_1, η_2 . This notation refers to fixed elements of the Cf., independently of the position they acquire in a particular cyclic arrangement. By contrast, e_h, e_i, η_h, η_i denote, for each arrangement, the elements that play the corresponding roles. The order of identification is determined by the order in which e_1, e_2, η_1, η_2 occur.

If one of e_1, e_2 is made the first vertex of the last polygon, Case IV occurs; otherwise one obtains Case I, II, or III according as the first two distinguished elements have different or equal indices. If e_1 and η_1 are neighboring, then e_2 and η_2 cannot also be neighboring, and an appropriate cyclic permutation gives Case I or II. The same holds symmetrically. If neither e_1 lies on η_1 nor e_2 on η_2 , then no two of the four elements are neighbors, and cyclic permutations allow an ordering that avoids Case IV and, unless two trial arrangements both yield Case III, directly gives Case I or II. In the exceptional ordering, making one of the points the second point of P_k gives Case II. Thus Case IV can always be reduced to Case I or II.

§16. The cubic curve

In Cases I and II the desired element e_i must be a point or tangent of a curve of order or class three determined by the previous construction.

For Case Ia, let the centers of the three pencils be $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$, and let the carriers of the three point rows be A, B, C , so that the pencil $\mathfrak{A}$ is projectively related to row A , and similarly for the other two. If p is a variable point of the plane and

$$p\mathfrak{A} = a, \quad p\mathfrak{B} = b, \quad p\mathfrak{C} = c,$$

and if a, b, c also denote the corresponding points on A, B, C , then we must find the points p for which a, b, c are collinear.

If p moves on a straight point row, then the rays a, b, c describe pencils and the points a, b, c describe point rows projectively related to each other. Thus one seeks triples of corresponding points on A, B, C that lie on one line. As p moves along the straight row, the line ab envelops a conic K , and the line ac envelops a conic K' . If a, b, c are collinear, their common line is a common tangent of the two conics. Apart from a , the two conics have at most three common tangents in general. Hence any straight line traced by p contains at most three points with the desired property; the locus is therefore a cubic curve.

If two points p_1, p_2 of this curve are known, then the line p_1p_2 contains a third point p_3 of the curve, which can be found linearly. This is reduced to constructing the fourth common tangent of two conics K, K' when three common tangents are known. Two of the known tangents are cut by the remaining tangents of each conic in projective point rows. Thus the points of one tangent are projectively related to those of the other in two ways. One double point is the intersection of the two known tangents; the other is obtained by a linear construction. The corresponding line is the fourth common tangent, and it determines the new point p_3 .

The corresponding lines $g = abc$ envelop a curve of class three. Just as two points of the cubic determine a third by a linear construction, two tangents of the class curve determine the third tangent through their intersection. The analogous constructions apply to Cases Ib, Ic, and Id. In Case II one obtains the same phenomenon from two pencils and two point rows: a moving point has an assigned line, and the desired points are precisely those that lie on their assigned lines. Once two such points on a line are known, the third can be found linearly.

The constructions presuppose that some elements of the curve are already known. This is so. In Case Ia the three centers $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$ lie on the required curve. In Case II, $\mathfrak{A}$ and $\mathfrak{B}$ lie on it, and a third point is obtained from the intersection of A and B ; the corresponding line gives a tangent of the class curve. The third points on the sides of the triangle $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$ are

$$(\mathfrak{B}\mathfrak{C}, A), \quad (C\mathfrak{A}, B), \quad (\mathfrak{A}\mathfrak{B}, C).$$

§17. Impossibility in Case III

The situation is quite different in the cases III. Considering the problem obtained in Case IIIc, one sees that construction of the required points by straightedge alone is impossible in general. This already follows from the fact that it can happen, and not merely in a special case, that no real system of such points exists at all.

Using the notation of §14, let $a, a_1; b, b_1; c, c_1$ be variable pairs of corresponding points on the six given lines. The lines aa_1 envelop a conic K , and the lines cc_1 envelop a conic K_1 . If the line b meets K and the line b_1 meets K_1 in real points, then each of these two lines is divided into two parts so that from one part real tangents to the associated conic are drawn, and from the other only imaginary tangents. If the projective correspondence between b and b_1 sends the segment of b belonging to real tangents entirely into the segment of b_1 belonging to imaginary tangents, then there is no pair of corresponding points b, b_1 from which real tangents go to both conics. This proves the assertion. The impossibility of a linear construction in Cases IIIa, IIIb, and IIIc is recognized in the same way.

§18. Conclusion

We have seen that every non-decomposing Cf. n_3 can be constructed by linear constructions with the addition of at most one quadratic construction. The construction given is completely general: every figure obtained by it is a Cf. of the required kind, and every Cf. of the required kind occurs among the figures obtainable by it.

For the second assertion, the construction must be understood with the special cases included. When two lines whose intersection is to be found coincide, an arbitrary point of that line is to be taken in place of the intersection; and when two points whose joining line is to be found coincide, an arbitrary line through them is to be taken.

Special cases must not be excluded. The problem of giving a general construction that produces all Cff. of the required kind in which distinct Cf.-points or Cf.-lines do not coincide is quite different from the one treated here. In many Cff. certain lines or points must in fact always coincide. One example is Schroeter's configuration

Cf. 10_3:

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1 1 1 2 2 3 3 4 5 8
2 4 6 4 5 6 7 6 7 9
3 5 7 8 9 8 9 10 10 10.
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If a Cf. n_3 is to be constructed linearly, then in decomposing it into polygons and arranging those polygons one must avoid Case III, since that case generally prevents the construction from

being completed by straightedge alone. Because the elements of the schema can be arranged in many ways so that each horizontal row contains every element once, because either pair of horizontal rows may be used to form the polygons, and because the order of the polygons and of the lines inside them can be chosen in several ways, it will usually be possible to produce one of Cases I and II. Whether this is possible for every Cf., or whether there are Cff. for which every arrangement leads to Case III, could not be decided.

In most Cff. it seems possible, by different arrangements of the elements, to produce any desired one of Cases I, II, III. In the Cf. 20₃ treated in §2, the three arrangements displayed in the German source arise from one and the same decomposition into the polygons

$$1, 9, 17, 16, 2, 15, 3, 10, 8; \quad 4, 7, 14, 11, 20, 6, 19, 12; \quad 5, 13, 18.$$

Cases I and II led to the problem of finding, from certain known points of a uniquely determined cubic curve, further points by linear construction, and similarly for the dual problem. In general it suffices to know one point of the curve in order to derive arbitrarily many other points from it, covering the whole curve or one of its branches so that the distances between successive points are smaller than any prescribed length. In special cases, however, even if three points of a uniquely determined nonsingular cubic are known, it is impossible by linear construction to determine any other point of the curve. Thus for each Cf. the applicability of the methods given for linear construction still requires a special examination.

Vita

I was born, Ernst Steinitz, in the village called Laurahuette on 13 June 1871, to my father Sigismund, whose death I deeply mourn, and to my mother Augusta, née Cohn. After attending the Friedrich Gymnasium in Breslau for ten years, I obtained the certificate of maturity in 1890. Soon thereafter I was admitted to the philosophical faculty and devoted eight semesters chiefly to mathematical studies, first in Breslau, then in Berlin, and then again in Breslau. I heard the distinguished teachers Baeumker, Dieterici, Hintze, Lipps, London, Meyer, Milch, Rosanes, Schmarsow, and Schroeter in Breslau; Aegidi, Delbrueck, Ebbinghaus, Frobenius, Fuchs, Grimm, Knoblauch, Kronecker, Planck, and Schwarz in Berlin. To all these men I offer my greatest thanks, above all to Jakob Rosanes, who, I gratefully acknowledge, supported my studies with the greatest goodwill.

Theses

1. The elements of number theory should already be taught at school.
2. The reintroduction of Latin as a written language would have many advantages for the mathematical sciences.
3. The geometry of n dimensions has no less reality as a mathematical discipline than the geometry of three dimensions.